COER UNIVERSITY
ROORKEE, UTTARAKHAND

7th, KM Haridwar, National Highway Vardhmanpuram,
Roorkee, Rehmadpur, Uttarakhand 247667

Lab Sheet-01

Python Programming Lab – Basic Operations with Arithmetic Operators and Control Statements

MCA I Semester (Session 2026-2027)

Objectives:

- To develop problem-solving skills using Python's arithmetic operators and control statements.
- To apply conditional logic for decision-making in basic programming tasks.

Requirements:

- Python 3.x installed on the system (IDLE, PyCharm, VS Code, or any preferred IDE).
- Basic understanding of Python syntax, variables, and input/output operations.
- Knowledge of arithmetic operators (+, -, *, /, %, **, //) and control statements (if, if-else, elif).

Program List:

1. Add two integers entered by the user.
2. Subtract two integers entered by the user.
3. Multiply two integers entered by the user.
4. Divide two integers and display the quotient and remainder.
5. Find the square and cube of a number.
6. Find the area of a rectangle.
7. Calculate the perimeter of a circle.
8. Swap two variables using a temporary variable.
9. Swap two variables without using a temporary variable.
10. Convert temperature from Celsius to Fahrenheit.

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11. Convert temperature from Fahrenheit to Celsius.
12. Calculate the simple interest given principal, rate, and time.
13. Calculate the compound interest given principal, rate, and time.
14. Calculate the average of three numbers.
15. Convert kilometers to miles.
16. Find the remainder when a number is divided by another number.
17. Calculate the power of a number without using pow().
18. Calculate the Body Mass Index (BMI).
19. Calculate the monthly EMI for a loan.
20. Find the percentage of marks obtained in 5 subjects.
21. Check if a number is positive or negative.
22. Check if a number is even or odd.
23. Check if a number is divisible by 5.
24. Check if a given year is a leap year.
25. Find the largest of two numbers.
26. Find the largest of three numbers.
27. Check if a character is a vowel or consonant.
28. Check if a character is uppercase, lowercase, or a digit.
29. Check if a number is in a given range.
30. Check if a number is prime.

Instructions:

- Write each program in a separate Python file named according to the question number (e.g., Q1_addition.py).
- Include comments in your code explaining the logic.
- Use proper variable names for better readability.
- Take input dynamically from the user (using input() function).
- Display output in a clear and formatted manner.

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Guidelines:

- Do not use built-in library functions unless specified in the problem statement.
- Test your program with different sets of inputs to ensure correctness.
- Follow proper indentation (4 spaces) for Python code blocks.
- Avoid hardcoding values; always use user input where applicable.
- Ensure programs run without syntax or runtime errors before submission.

Submission:

- Submit the source code files (.py) for each task along with a brief report documenting the implementation details and the results of the executions.